

1888. Minimum Number of Flips to Make the Binary String Alternating

Solved 

Medium

 Topics

 Companies

 Hint

You are given a binary string `s`. You are allowed to perform two types of operations on the string in any sequence:

- **Type-1: Remove** the character at the start of the string `s` and **append** it to the end of the string.
- **Type-2: Pick** any character in `s` and **flip** its value, i.e., if its value is `'0'` it becomes `'1'` and vice-versa.

Return *the **minimum** number of **type-2** operations you need to perform such that `s` becomes **alternating**.*

The string is called **alternating** if no two adjacent characters are equal.

- For example, the strings `"010"` and `"1010"` are alternating, while the string `"0100"` is not.

Example 1:

Input: `s = "111000"`

Output: `2`

Explanation: Use the first operation two times to make `s = "100011"`.

Then, use the second operation on the third and sixth elements to make `s = "101010"`.

Example 2:

Input: `s = "010"`

Output: `0`

Explanation: The string is already alternating.

Example 3:

Input: `s = "1110"`

Output: `1`

Explanation: Use the second operation on the second element to make `s = "1010"`.

Constraints:

- `1 <= s.length <= 105`
- `s[i]` is either `'0'` or `'1'`.

Python:

class Solution:

```
def minFlips(self, s):
    n = len(s)
    e, o = (n + 1) // 2, n // 2
    x = s[::2].count('1') - s[1::2].count('1')
    r = min(e - x, o + x)
    if n & 1:
        for c in s:
            x = 2 * (ord(c) & 1) - x
            r = min(r, e - x, o + x)
    return r
```

JavaScript:

```
var minFlips = function(s) {
    let n = s.length;
    let ss = s + s;

    let ans = Infinity;
    let diff1 = 0, diff2 = 0;

    for(let i = 0; i < ss.length; i++){

        let exp1 = (i % 2 === 0) ? '0' : '1';
        let exp2 = (i % 2 === 0) ? '1' : '0';

        if(ss[i] !== exp1) diff1++;
        if(ss[i] !== exp2) diff2++;

        if(i >= n){
            let p1 = ((i-n) % 2 === 0) ? '0' : '1';
            let p2 = ((i-n) % 2 === 0) ? '1' : '0';

            if(ss[i-n] !== p1) diff1--;
            if(ss[i-n] !== p2) diff2--;
        }

        if(i >= n-1){
            ans = Math.min(ans, diff1, diff2);
        }
    }

    return ans;
};
```

Java:

```
class Solution {
    public static int minFlips(String s) {
        final int n = s.length();
        int p = 0, k = 0;
        for (int i = 0; i < n; i++) {
            final char c = s.charAt(i);
            k += p ^ c & 1;
            p ^= 1;
        }
        int r = Math.min(k, n - k);
        if ((n & 1) != 0) {
```

```
p = 0;
for (int i = 0; i + 1 < n; i++) {
    final char c = s.charAt(i);
    k += (p ^ c & 1 ^ 1) - (p ^ c & 1);
    p ^= 1;
    r = Math.min(r, Math.min(k, n - k));
}
return r;
}
```